Quantum-Safe Dynamic Data Auditing with Batch Verification for Blockchain-Enabled Edge-IoT Networks

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Abstract

1 Introduction

2 Results

All experiments and evaluations were conducted on a local workstation running the Windows 11 operating system. The hardware configuration consisted of an Intel Core i7 (13th Generation) processor with a base clock speed of 2.6 GHz, coupled with 16 GB of system RAM. For tasks requiring hardware acceleration, the system utilized a dedicated NVIDIA GeForce RTX 4050 graphics processing unit (GPU) equipped with 6 GB of VRAM. The software environment was established using Python version 3.13.5, with core numerical computations and array manipulations performed using the NumPy library, version 2.4.3.

Table 1 Communication costs for state-of-the-art

Sr. N	Schemes	Public Parameter Size	Secret Key Size	Proof/Ciphertext Size	Public Key Size
1	[1]	$(3NM+N)\|Z_p\|$	-	$2M\|Z_p\|$	$(2M+1)\|Z_p\|$
2	[2]	$(3NM+N)\|Z_p\|$	-	$2M\|Z_p\|$	$(2M+1)\|Z_p\|$
3	[3]	$(NM+N^2)\|Z_p\|$	-	$2M\|Z_p\|$	$(3M+1)\|Z_p\|$
4	[4]	$(NM+N)\|Z_p\|$	-	$M\|Z_p\|$	$(2M+1)\|Z_p\|$
5	[5]	$MN\|Z_p\|$	$M^2\|Z_p\|$	$NM\|Z_p\|$	$(3M+1)\|Z_p\|$
6	[6]	$MN\|Z_p\|$	$M^2\|Z_p\|$	$2M\|Z_p\|$	$(M+N)\|Z_p\|$
7	[7]	$MN\|Z_p\|$	$M^2\|Z_p\|$	$M\|Z_p\|$	$(2M+1)\|Z_p\|$
8	Proposed	$NM\|Z_p\|$	$M\|Z_p\|$	$N\|Z_p\|$	$N\|Z_p\|$

Our scheme employs single-vector Module-LWE combined with Lyubashevsky's rejection sampling, which ensures zero-knowledge properties without requiring matrix-based secret keys. Since the one way function $t = A \cdot s \pmod p$ acheives security with a single vector $s \in \mathbb{Z}_p^M$ and the public matrix $A \in \mathbb{Z}_p^{N\times M}$ provides sufficient entropy for lattice hardness, the full M^2 matrix structure used in traditional schemes is unnecessary. Rejection sampling eliminates the need for error correcting codes that would require matrix dependency, allowing us to reduce secret key size from $M^2|\mathbb{Z}_p|$ to $M|\mathbb{Z}_p|$. Proof size is reduced as the only {msg, challenge, and response} is sent as the Zero-Knowledge proof.

Table 2 Execution Time for Cryptographic Primitive Operations

Symbol	**Operation**	**Time (μs)**
T_H	Hash operation (SHAKE-256) in $\mathbb{Z}_q^N$	12.80
T_G	Uniform sampling from $[-\eta,\eta]^M$	13.84
T_+	Vector addition or scalar-vector multiplication mod q	3.82
T_*	Matrix-vector multiplication mod q	366.86

Table 3 Comparison of Computational Costs

Scheme	**Sign/Proof Generation Cost**	**Verification Cost**
[8]	$T_H + 4T_G + 6T_+ + 6T_*$	$T_H + 2T_G + 5T_+ + 5T_*$
[9]	$2T_H + T_G + 5T_+ + 5T_*$	$2T_H + 2T_+ + 4T_*$
[10]	$3T_H + T_G + 4T_+ + 4T_*$	$T_H + T_G + 2T_+ + 2T_*$
[11]	$3T_H + 7T_+ + 5T_*$	$T_H + 5T_+ + 5T_*$
[12]	$3T_H + 6T_+ + 6T_*$	$T_H + 4T_+ + 4T_*$
[13]	$2T_H + 2T_G + 2T_+ + 4T_*$	$T_H + 2T_G + 2T_+ + 2T_*$
[14]	$2T_H + 4T_+ + 4T_*$	$T_H + 2T_+ + 3T_*$
Proposed	$\mathbf{T_H + T_G + 3T_+ + T_*}$	$\mathbf{T_H + 2T_+ + T_*}$

For proof generation cost, the prover performs one sampling step which takes T_G time, computes commitment $w = A \cdot y \pmod p$ which takes T_* time, computes challenge $c = H(w\|m)$ which takes T_H time and finally computes response $z = y + c \cdot s$ which takes approximately $3T_+$ time. The key reason for lower cost is that our scheme avoids repeated commitments/hashes and keeps the xpensive lattice multiplication to a single instance per proof, while the remaining computations are low-cost modular arithmetic.

For proof verification time, the verifier recomputes Az which takes T_* time, computes and combines ct and $w' = Az - ct \pmod q$ which takes $2T_+$ time, and finally computes $c' = H(w'\|m)$ which takes T_H time to compare with the original challenge c.

Table 4 Batch Verification Efficiency Analysis

Batch Size (N)	**Individual Total** (μs)	**Batch Total** (μs)	**Time per Tx** (μs)	**Speedup Factor**
25	9,682.50	1,359.18	54.37	7.12×
50	19,365.00	2,347.68	46.95	8.25×
75	29,047.50	3,336.18	44.48	8.71×
100	38,730.00	4,324.68	43.25	8.95×

Individual: $N \times T_{\text{verify}}$ where $T_{\text{verify}} = T_* + 2T_+ + T_H = 387.30\ \mu s$
Batch: $N \cdot T_H + 7N \cdot T_+ + T_* + T_+$

In batch verification, the verifier hashes each proof once and accumulates weighted terms across all proofs, then performs only one global matrix-vector multiplication for the entire batch and one final modular check. This gives the batch cost $T_{batch} = N \cdot T_H + 7NT_+ + T_* + T_+$, where N is the batch size. The key efficiency point is that the dominant lattice multiplication term T_* is paid once per batch, not once per proof, so per-transaction verification time decreases as N grows.

Table 5 Theoretical Formula vs. Simulated Results Validation

Operation	**Formula**	**Theoretical** (μs)	**Simulated** (μs)	**Ratio**
Key Generation	$T_G + T_*$	380.70	506.93	0.75
Proof Generation	$(1+r)(T_G+T_*+T_H+3T_+)$	438.62	571.29	0.77
Single Verification	$T_* + 2T_+ + T_H$	387.30	354.29	1.09
Batch Verification (N=100)	$N{\cdot}T_H + 7N{\cdot}T_+ + T_* + T_+$	4324.68	3457.55	1.25

$r = 0.083$ (measured rejection rate). Ratio = Theoretical/Simulated.
Primitives: T_H=12.80μs, T_G=13.84μs, T_+=3.82μs, T_*=366.86μs.

Table 6 Batch Verification Speedup: Theoretical vs. Simulated

Batch Size	**Theoretical**		**Simulated**	
(N)	**Time** (μs)	**Speedup**	**Time** (μs)	**Speedup**
25	1,359.18	7.12×	1,172.81	8.25×
50	2,347.68	8.25×	2,106.90	9.19×
75	3,336.18	8.71×	2,976.18	9.76×
100	4,324.68	8.95×	3,457.55	11.20×

Theoretical speedup = $\frac{N \times T_{verify}}{T_{batch}}$.

As shown in Fig.1, the verifier performs only one global matrix-vector multiplication for the entire batch ($AZ = A \cdot Z_{\text{batch}}$) and then checks a single aggregate equation, while per-proof work is limited to hashing and modular vector updates. Hence, the batch latency follows

$$T_{\text{batch}}(N) = T_* + N(T_H + 7T_+) + T_+,$$

which is affine in N with a small slope. Therefore, the practical behavior appears near-constant over the tested range ($25 \leq N \leq 100$), and the amortized per-transaction cost

$$\frac{T_{\text{batch}}(N)}{N} = (T_H + 7T_+) + \frac{T_* + T_+}{N}$$

decreases as N increases. This is why the proposed curve grows slowly compared with conventional linear verification schemes.

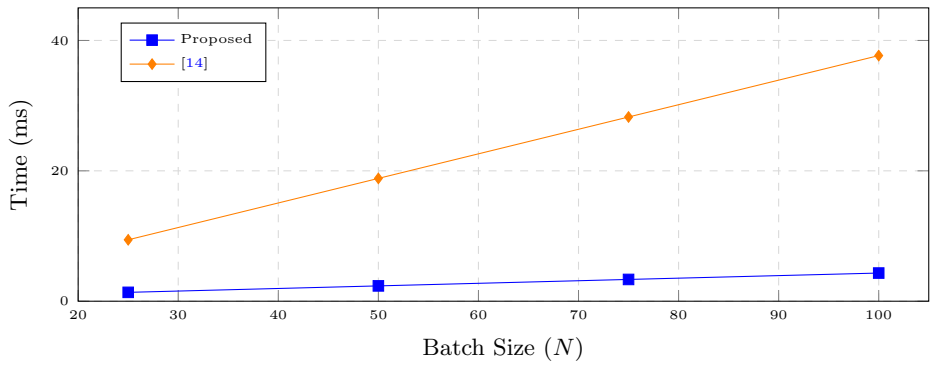

Fig. 1 Batch Verification Latency

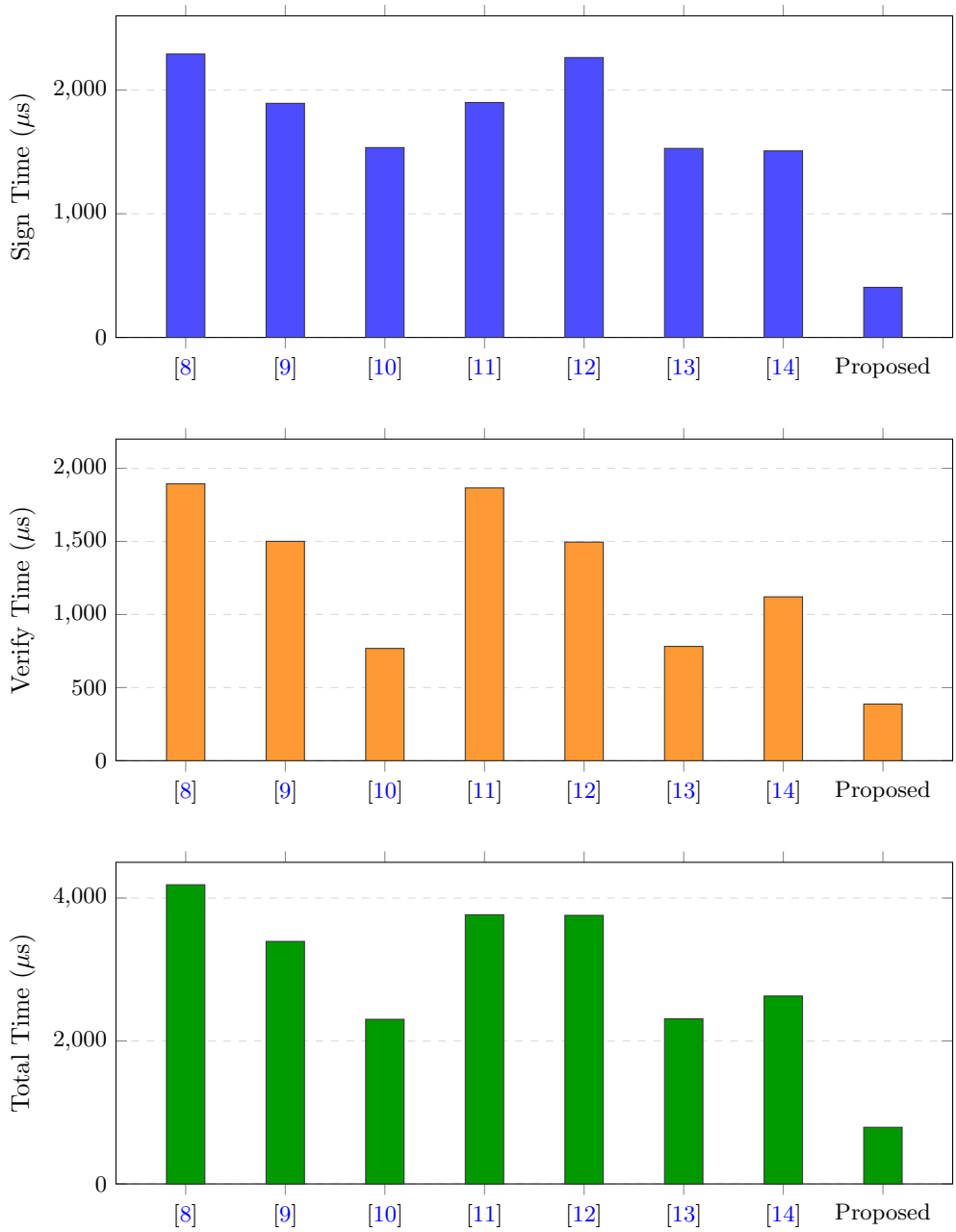

Fig. 2 Performance Comparison: Individual metrics for Sign, Verify, and Total Execution Times (μs).

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